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ORIGINAL ARTICLES

Perspectives on AI literacy in Middle School Classrooms: An Integrative Review

Katarina Sperling[1] · Linnéa Stenliden[1] · Linda Mannila[2] · Jonas Hallström[1] · Charlotta Nordlöf[1] · Fredrik Heintz[2]

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Abstract

AI literacy in school education is booming within the scientific discourse of AI in education. How AI literacy is currently being framed serves diverse educational, political, and commercial purposes influencing how we imagine postdigital classrooms today and in the future. More importantly, how AI literacy emerges in primary education notably impacts how children understand AI and their own agency in a society where AI is ubiquitous. This study reviews how scientific literature conceptualises AI literacy, focusing on middle school students. An AI-adapted literacy framework (GeST) is used in the analysis to distinguish three perspectives of AI literacy (Generic, Situated, and Transformative). Forty-four papers from 2016–2024 were included in the final descriptive and qualitative analysis, showing an exponential growth in scientific papers. While still vaguely defined and poorly theorised, AI literacy materialises into different AI curricula and technology-supported teaching activities. The GeST analysis indicates that AI literacy is primarily viewed as a set of measurable skills related to generalisable theoretical knowledge that is expected to make children more competitive in a globalised and technologised world. Although some papers consider empowering students with specific competencies to challenge the AI development, critical considerations of AI in education is less visible. The paper highlights the necessity to steer the conceptualisation of AI literacy to put a stronger emphasis on critical orientations that enable students as well as teachers to examine claims about AI, and pose ethical questions to its adoption and use in classrooms and beyond.

Keywords AI literacy · Middle school · Primary education · Postdigital · K-12 classroom

✉ Katarina Sperling
katarina.sperling@liu.se

[1] Department of Behavioural Sciences and Learning, Linköping University, Linköpings Universitet, 601 74 Norrköping, Sweden

[2] Department of Computer and Information Science, Linköping University, Linköping, Sweden

Introduction

The presence of digital (AI) technology through e.g. recommender systems, large language models, and affective computing has seen a boom in the lives of middle school students, shaping not only their decisions, worldviews, and agency but also their educational experiences (Lupton and Williamson 2017; Smith et al. 2024). Many middle schoolers engage with AI-driven social media platforms that use AI to curate content, influencing students' future visions of the self and society. These students are the first generation whose education, healthcare, and spare time are increasingly mediated by AI-powered applications (UNICEF 2021). In parallel, different AI technologies are being introduced in schools to monitor students, manage groups, automate grading, and deliver data-driven instructions through AI chatbots and intelligent tutoring systems, raising various ethical concerns related to privacy and security, the risks of bias and fairness, transparency, and teacher and student relations (Crompton and Burke 2022; Selwyn et al. 2023; Stenliden and Sperling 2024). The key problematic is how to best prepare this generation, in their transformation from children to lifelong learners, with sound and appropriate knowledge about AI and its effects. In response, a call for enhancing young learners' understanding of what AI is and providing them with skills to evaluate and critically engage with AI technologies, their *AI literacy*, has grown in importance within the educational discourse, as evidenced by its increasing presence in policy reports (Holmes et al. 2022) and scientific literature (Ng et al. 2021; Chiu et al. 2024). Here, AI literacy emerges as a new form of civic knowledge, proposed as a solution to diverse challenges in postdigital societies (cf. Rahm 2024). As such, research on AI literacy is both shaped by and shapes global and national politics, enacting different perspectives and powerful visions of *postdigital classrooms* (Forsler et al. 2024; Lindberg and Johansson 2023). In these classrooms educational technologies play a crucial role in shaping how the future of schooling, in this study centred around middle school education, is imagined – both as material artefacts and as subjects or *literacies* that must be taught (Mannila et al. 2025). Here, technologies are integrated into and instrumental for teaching and learning practices (Jandrić and Knox 2022). Their ubiquitous presence is also evident in contemporary and emergent curricula and expected knowledge and skills (Aitken and Jones 2023). This warrants comprehensive reviews of how AI literacy is conceptualised.

Several literature reviews have synthesised what AI literacy entails in various domains, including K-12 education (Casal-Otero et al. 2023; Eguchi 2021; Kandlhofer et al. 2016; Ng et al. 2022a; Stolpe and Hallström 2024), early childhood education (Su et al. 2023), higher education (Bearman et al. 2023; Laupichler et al. 2022), the workplace (Cetindamar et al. 2022), as well as among the general public (Long et al. 2022; Long and Magerko 2020; Schüller 2022). The academic interest for AI literacy in schools is also reflected in the increasing number of teaching and learning digital resources available for K-12 (Druga et al. 2022; Touretzky et al. 2022). Taken together, these studies highlight a rapidly expanding research field that is growing in both size and influence, with many studies

shaped by computer science and exploratory teaching approaches (Sperling et al. 2024). A recent study by Velander and colleagues (2024) pointed out that current definitions of AI literacy often fall short in suggesting how this understanding may facilitate learners to examine AI and its impact critically. However, to our knowledge no study exists that looks at different perspectives and rationales behind what and why specific age groups should acquire knowledge about AI and the potential outcomes they could achieve with such knowledge. As previously argued, middle school students represent an age group where it is essential to cultivate ethical, critical, and creative engagements with AI technologies. This integrative review aims to address this gap by critically exploring the current state and perspectives of AI literacy in the scientific literature focusing on middle school students (9 to 13-year-olds). We address this aim through the following research questions:

1. What is the current state of AI literacy relevant for 9 to 13-year-old-students?
2. What perspectives of AI literacy emerge in the literature?

To critically explore AI literacy, we begin with a brief overview of the literacy concept and its relationship to AI. Next, we introduce the selected framework used to define AI literacy and outline our review methodology. The findings are presented in two stages: first, a brief descriptive overview followed by an analytical examination of how AI literacy is emerging. We conclude by discussing the implications of how AI literacy, as an element of the future making of middle school classrooms, is currently conceptualised and propose some avenues for future research.

The Emergence of AI Literacy

The increasing presence of digital technologies in everyday life has shifted the traditional concept of literacy, expanding it from simply the ability to read and write to encompass a broader range of skills and competencies (Cazden et al. 1996; Jewitt 2008; Kress 2003; Leu et al. 2017; Serafini 2014; Unsworth 2014). In response, a variety of literacies have surfaced, each focused on specific technological domains, such as digital literacy (Bawden 2001), computational literacy (DiSessa 2018), critical digital literacies (Ilomäki et al. 2023), media literacy (Potter 2018), information literacy (Tuominen et al. 2005), data literacy (Wolff et al. 2016), visual literacy (Bodén and Stenliden 2019), visualisation literacy (Firat 2022), and algorithm literacy (Ridley and Pawlick-Potts 2021). This intricate landscape of literacies mirrors the close integration of literacy with emerging technologies as part of postdigital meaning-making (Bhatt and Gourlay 2024), as well as the increasing demand for citizens to stay up to date with the latest technological advancements. AI literacy, which is in focus in this paper, can still be considered a new literacy concept rooted in the computer science domain (Sperling et al. 2024) and with strong connections to data literacy (Schüller 2022). These connections are particularly visible in various national AI curricula, which take on a technological perspective by covering AI

technologies and applications, including techniques like machine learning and natural language processing (Miao and Shiohira 2022). At the same time, it is argued that AI technologies often operate autonomously and can adapt to their context, enabling them to make decisions and take actions based on predictions. Therefore, understanding the role of AI ethics and the potential impact on employment and sustainable development have been identified as pivotal parts of AI literacy (Holmes et al. 2022).

While there is as yet no universally accepted definition of AI literacy, a highly cited definition within and outside of scientific literature is that of Long and Magerko (2020:2): 'a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace'. This definition tends to emphasise the individual dimension of a literacy that one can or cannot have. However, AI literacy could also be seen as something emerging within social contexts and communities of practices (Hawkins 2013; Lave & Wenger, 1991). Understanding literacy in this way requires a comprehensive approach that considers general as well as specific knowledge domains, situated organisational contexts, and practical tasks (Tuominen et al. 2005). Beyond mere proficiency, this interconnected view also includes critical perspectives on literacy as a tool for navigating power dynamics and fostering empowerment (Delgado-Gaitan 2013; Freire 1972).

Accordingly, we argue that AI literacy emerges from the many complex relationships between social, material and semiotic resources (Burnett & Merchant, 2020), serves diverse strategic purposes in different political and organisational contexts (Selwyn 2024), and reflects a *postdigital condition* (Bhatt 2023: 3) in which digital technologies also underpin literacy practices in classrooms. The way AI literacy is conceptualised has implications for how we imagine postdigital classrooms, as well as how young people understand AI and their agency in a society where AI is ubiquitous. Based on these standpoints, we take inspiration from the Generic, Situated, and Transformative (GeST) framework originally developed for information literacy by Lupton and Bruce (2010). When adapted to this study, we found it productive in offering a comprehensive way to map different perspectives on how AI literacy is presented in the scientific literature relevant to middle school students.

The GeST Framework

The GeST framework draws on many existing literacy models and theories and offers distinct conceptualisations of literacy which can be adapted to other literacies such as AI literacy. Thus, we have found the framework useful as a way of mapping and analysing scientific literature on a still under-theorised AI literacy. Moreover, AI literacy is interlinked with information literacy, for example when it comes to the ability to evaluate AI and its outputs critically (Jandrić 2019). The GeST framework approaches literacy from three interlinked perspectives: (1) as a set of *generic* and individual skills, (2) as *situated* in social practices, or (3) as *transformative* for both individuals and society (Lupton and Bruce 2010). Broadly speaking, the generic perspective captures a more behaviourally oriented form of literacy, the situated

perspective emphasises sociocultural dimensions, and the transformative perspective focuses critically on emancipation. At the same time, the different GeST perspectives form a hierarchy of increasing complexity where the *generic* perspective can be seen as the most basic, and the *transformative* as more complex. This is not to suggest that the different perspectives are equivalent to different stages of maturation. As stated by Lupton and Bruce (2010:6), 'in learning contexts, a hierarchy suggests that curricula should attend to the full complexity of the literacy experience'.

A Generic Literacy

The first perspective, the *generic*, sees literacy as a set of external and useful skills that can be learned by individuals, observed and measured. Such skills can include theoretical understanding of specific phenomena. In the context of AI, this could entail how data are used to develop machine learning algorithms or how various AI-based technologies for generating text, images, and sounds are applied in different societal realms. Within this perspective, literacy is often linked to skills that underpin a nation's development and knowledge advancement. This knowledge is viewed as crucial in fostering a flexible workforce capable of competing in a globalised and technologically advanced world.

A Situated Literacy

The second, *situated*, perspective views literacy as an authentic and fundamentally collective activity that emerges in specific settings. From this outset, literacy is seen as something social, and literacy practices as always inherently contextually bound activities. Thus, there is not one universal literacy but multiple literacies, which in turn means that AI knowledge or skills cannot simply be transferred from one context to another. To measure or assess literacy, it is necessary to understand what it means both for the individual *and* for the community in which it emerges.

A Transformative Literacy

The third, *transformative*, perspective builds on the situated dimension by adding emancipatory outcomes. This perspective necessitates having a critical and conscious mindset, asking for what purpose, in whose interest, and what the implicit and explicit assumptions are. Lupton and Bruce (2010) consider the transformative perspective as the most inclusive, as it merges the outcomes of the generic and situated aspects by emphasising the outcomes for the individual and society. This stands in contrast to a generic perspective that focuses solely on skills within an individual, and the situated perspective focusing solely on context and collaboration.

How GeST is Applied in this Study

To further highlight how the three different perspectives emerge in the scientific literature, we selected and adapted three propositions out of 10 previously identified

by Lupton and Bruce (2010:15) in relation to information literacy: (1) AI literacy is …, (2) AI literacy is important because …, and (3) AI literacy is being taught and learnt through …. These propositions were chosen because they (1) capture general assumptions about AI literacy, (2) reflect the underlying rationale of the various studies, and (3) reveal assumptions about schools, students, teachers, and pedagogy. By applying these different propositions, we were able to more clearly and comprehensively identify the presence of generic, situated, and transformative perspectives, and to determine whether each paper represented one or multiple perspectives. A more detailed description of the characteristics of each proposition in relation to each perspective and how they were used to map the literature is provided in the Method section and summarised in Table 2. While our analysis set out to distinguish different perspectives, it is important to underline that the GeST framework acknowledges all three dimensions as instrumental and interdependent, and does not favour any specific perspective. For example, to critically reflect on AI in education, one needs both generalisable knowledge and context-based practical knowledge and skills. Similarly, for AI literacy to lead to empowerment, it must be viewed as a social and collective endeavour.

Methodology

We conducted an integrative review to evaluate the literature in a manner that could both advance the knowledge of AI literacy in the context of middle school education and generate novel approaches and frameworks (Snyder 2019). An integrative review does not follow specific standards and is commonly applied to address emerging topics and to assess, critique and synthesise the literature for new theoretical perspectives and dimensions (Torraco 2005). In line with Whittemore and Knafl's recommendation (2005), the approach has been used to critically address and examine the main ideas and relationships of the literature in a transparent and structured way, which will be outlined next.

Search Strategy and Selection

The search was conducted on 18 February 2025, using Scopus, a well-recognised bibliometric database known for its comprehensive coverage of peer-reviewed, high-quality journal articles, conference papers, and book chapters. Given the aim of this study, which is to critically explore the current state and perspectives of AI literacy in the scientific literature focusing on middle school students, we determined that Scopus alone yielded relevant papers that could offer a comprehensive overview of the current discourse. This conclusion was drawn based on test searches conducted in both Scopus and Web of Science. We used the search query ("AI literacy"OR"artificial intelligence literacy") AND (K-12 OR middle OR primary) to ensure that studies focusing on middle school students were captured, even when the educational level was described more broadly, such as with the term K-12. We restricted our search to include only papers published up to and including 2024.

Additionally, we limited the document types to published conference papers, journal articles, and book chapters to ensure the literature had undergone a peer-review process. Furthermore, we focused only on publications in English. All the identified items (n = 141) were imported into a reference management software (Zotero) and submitted for screening. We then conducted a two-stage screening process. First, two of the authors read the abstracts of all 141 records and excluded studies based on selection criteria one to seven. To extract relevant information from the studies, a data charting table was created, including information about the study, main topic, and its relevance to the GeST framework.

During the second stage in the screening process, we read the papers more comprehensively and decided to add exclusion criteria eight and nine, and exclude papers that were not full studies and papers that referenced the same study but at distinct stages in the research process. For example, a conference paper in 2021, which was considered relevant given the scope of the review, was removed as the same study was later elaborated upon in a scientific paper by the same group of authors. Thus, we decided to keep only the more developed version (Table 1).

Based on the outcomes of this meticulous second screening, 44 articles were chosen for a final analysis (Fig. 1).

Analysis

During the analysis, the 44 papers underwent a closer reading in relation to the GeST framework. First, we categorised all papers according to a more general description of the three perspectives (generic, situated, and transformative). As we soon noticed that several perspectives could often be found within one paper but concerning different aspects, we decided to select and adapt three propositions inspired by the GeST framework (AI literacy is …, AI literacy is important because …, and AI literacy is being taught and learned through …) so that one paper could have three potential mappings within each proposition (Table 2). With these three propositions in mind, we revisited all papers and mapped each one according to the generic, situated, and transformative perspective.

Although this process required a more meticulous reading, it added greater nuance to the analysis. Where studies expressed more than one perspective

Table 1 Inclusion and exclusion criteria

Include	Exclude
1. In English	1. All languages apart from English
2. Peer reviewed studies (journals, conference proceedings and book chapters)	2. Grey literature
3. Focus on AI literacy	3. Unpublished articles
4. AI literacy in school classrooms	4. No focus on AI literacy
5. Middle school student focus	5. AI literacy outside of school classrooms
	6. No middle school student focus
	7. Literature reviews
	8. Proof-of-concept papers
	9. Duplication of research study

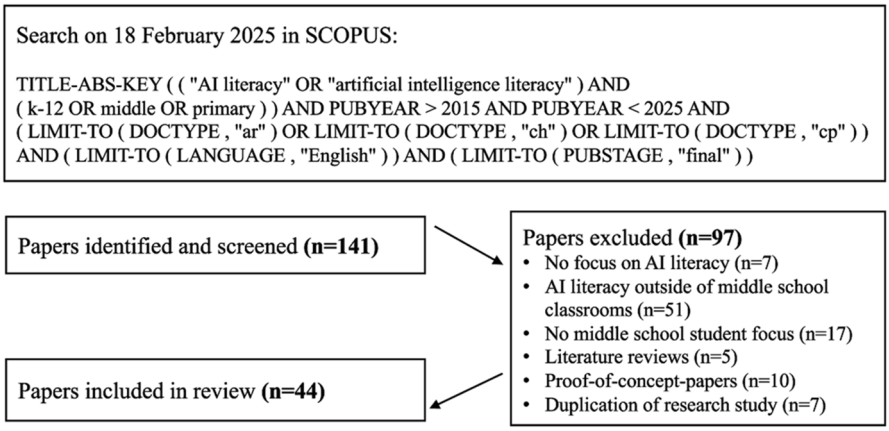


Fig. 1 Study selection

concerning the same proposition, we marked them all. In cases where we could not discern any stance related to a proposition, we marked it as n/a. This resulted in some papers reflecting only one perspective in relation to the three propositions, while other papers reflected all three perspectives. For example, a paper presenting AI as a set of skills that can be transferred from one context to another, also aiming to equip teachers with teaching methods that empowered students, reflected a conceptualisation of AI literacy as both generic and transformative. In cases where both a situated and a transformative perspective emerged within the same proposition, we chose the one we considered to be the most prominent.

Limitations

Whilst this integrative review was undertaken as rigorously as possible, it is by necessity limited by its search strategy, database choice, and inclusion and exclusion criteria. Nevertheless, our primary focus in this paper was to methodologically advance and critically explore AI literacy in middle school classrooms by adding new perspectives on this emerging research field. We also recognise the limitations of our chosen framework, which may have overlooked other dimensions presented in the papers.

Findings

A total of 44 papers serves as the basis for the analysis. In response to our first research question, we describe the current state of AI literacy relevant to middle school students as stated in the literature. This section encompasses background details, including publication year and country of origin. It also elicits the predominant topics and how AI literacy is framed within the papers. Next, and in response to

Table 2 Adapted GeST propositions used in the analysis

Proposition	Generic	Situated	Transformative
AI literacy is …	… a set of cognitive skills and theoretical knowledge about e.g. training data, how machine learning works, bias, deep fakes, and an understanding of how the technology is implemented and used in different areas of application, e.g. in middle school students' learning and everyday life	… a range of situated AI practises and knowledge adapted to specific subjects, ages, school communities, and culture, and taught in diverse ways	… a range of situated AI practises used to transform oneself and society, consisting of the implicit and explicit meanings and assumptions inherent in educational practices
AI literacy is important because …	… AI is ubiquitous in society and middle school students need to know how AI works. This also requires a flexible workforce to remain competitive in a globalised, technology-driven world, while also having the potential to make teaching and learning more efficient and effective	… middle school students need AI competencies to understand how AI affects society in specific contexts and to interact with AI, create new knowledge, and solve real-life problems in relation to specific contexts	… middle school students need to be empowered to challenge the status quo of the use and development of AI in society
AI literacy is taught and learned through …	… AI curricula and generic learning material about AI that middle school students should engage/interact with to understand the basics of AI and its technological foundations through e.g. hands-on activities regarding training data and bias	… teaching activities in contextualised settings where middle school students engage in teaching activities with authentic examples of AI use, e.g. the social, historical, political, and economic context of AI	…teaching activities empowering middle school students to engage in AI practices, e.g. by examining whose interests are served, who is silent, and how AI is designed and developed

our second research question, we present how the generic, situated, and transformative perspectives emerge.

RQ1: Current State of AI Literacy

AI literacy is growing in interest in the scientific community, with almost half of the included contributions (n = 21) published in 2024 and with a relatively even distribution between conference proceedings and journal papers (Fig. 2). This illustrates the increasing research interest in AI literacy. Proceedings primarily stem from conferences related to computer science education, artificial intelligence, educational technology, and technology-enhanced learning (TEL). The most frequently featured journals include *Computers and Education: Artificial Intelligence* (n = 8), *International Journal of Artificial Intelligence in Education* (n = 3), and *Künstliche Intelligenz* (n = 2).

AI in the context of middle school education is a topic of global interest, with almost half of the included studies (n = 19) originating from the United States and around one-fifth (n = 9) from China. Other countries with researchers contributing to this discussion include the UK (n = 3), Germany (n = 2), Spain (n = 2), Finland (n = 2) and one each from Austria, Denmark, South Korea, Canada, Belgium, Malta, and Greece.

During our reading we identified three overarching themes: 1. Teaching materials/activities (n = 19) – various interventions with students which aim to develop digital and analogue artefacts and activities for teaching with and about AI in middle school classrooms; 2. Curriculum development (n = 19) – the design and development of curricula, frameworks, and principles which define what middle school students should know about AI; and 3. Assessment of student knowledge (n = 6) – studies which examine what middle school students think and know about AI.

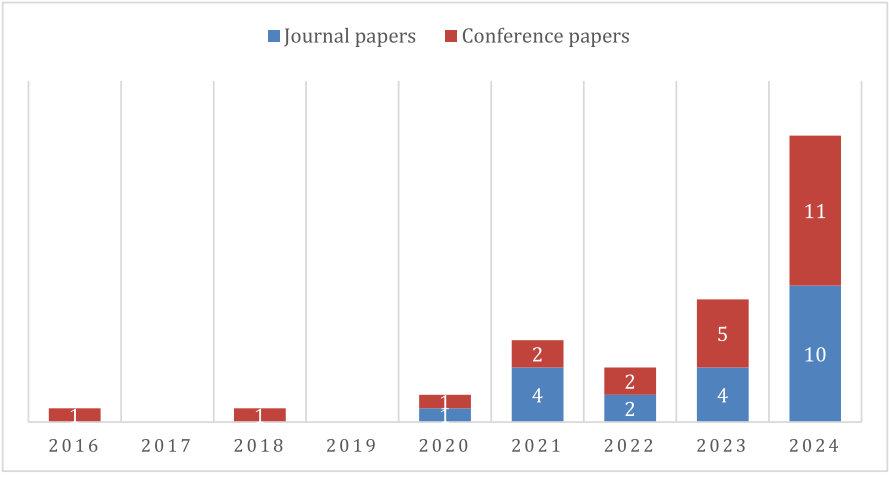


Fig. 2 Number of conference papers and journal articles on AI literacy in middle school between 2016 and 2024

While the term *AI literacy* appears in all the papers, in most cases it is used in a superficial manner to advocate for specific teaching activities or curriculum development. Definitions of AI are even rarer, often taking for granted either the concept of AI itself or the notion of human intelligence. The ways in which AI literacy is defined, operationalised, and measured vary significantly, and only a few papers provide a detailed explanation of what AI literacy entails. For instance, in Zhang et al. (2024), a section states that 'AI literacy has progressively expanded to encompass various principles, skills, types of knowledge, and multiple academic domains', citing Long and Magerko (2020), an influential seminal paper summarising what was then considered as necessary understanding of AI and how it should be taught.

RQ2: Perspectives of AI Literacy

In response to RQ2, the AI-adapted literacy framework (GeST) was used to identify generic, situated, and transformative perspectives based on three propositions (Table 3). Our mapping shows that AI literacy is predominantly presented from a generic perspective in the literature, as reflected in 79 occurrences across all three propositions. Thus, on a general level, AI literacy, in the context of middle school classrooms, emerges as a set of measurable skills and content knowledge that an individual can possess to varying degrees. These skills are presented as something that can be learned, most commonly through hands-on activities and digital artefacts (e.g. games, robots, conversational agents, and AI systems), and assessed and measured through different surveys and tests.

The other perspectives were less prominent in the included articles: the situated one was reflected 21 times across the three propositions, and the transformative, 23 times. These perspectives advocate for the introduction of AI from an emancipatory and critical perspective, emphasising ethics and the context and culture within which teaching and learning happens. Although they tend to overlap (Lupton and Bruce 2010), these perspectives are, even when added together (for a total of 44 times), much less evident in the literature. It is noteworthy that not all propositions were reflected in all papers and that a majority of the articles reflect more than one perspective. The next sections will exemplify how the different perspectives on AI literacy emerged in relation to the three different propositions: AI literacy is …, AI literacy is important because …, and AI literacy is taught and learnt through …. (Table 4).

AI literacy is …

The first proposition – how AI literacy is conceptualised – is predominantly shaped by a generic perspective, as reflected in Table 3, with about four-fifths of the scientific literature (n = 35) aligning with this view. Here AI literacy is equivalent to discrete skills and knowledge about, for example, machine learning, generative AI, deep fake, or AI ethics. These are skills and knowledge that middle school students may possess to different extents which researchers can observe and measure using statistical methods like pre- and post-tests. This perspective emerges either implicitly or is stated in studies

Table 3 Overarching themes in the literature

Nr	Topics	Studies
1	Teaching materials/activities (n = 19)	Ali et al. 2021, 2023; Bilstrup et al. 2022; Breideband et al. 2023; Ng et al. 2022b; Drivas and Doukakis 2022; Eguchi 2023; Henry et al. 2021; Kahila et al. 2024; Kong et al. 2023; Pearce et al. 2023; Petridis et al. 2024; Schaper et al. 2020; Tseng et al. 2024; Voulgari et al. 2021; Williams et al. 2024; Yu and Huang 2024; Zhou et al. 2024a; Zhou et al. 2024b
2	Curriculum development (n = 19)	Burriss et al. 2024; Chiu et al. 2024; Choi et al. 2024; DiPaola et al. 2024; Hollands et al. 2024; Höper and Schulte 2024; Kim and An 2024; Moreno-León et al. 2024; Simbeck and Kalff 2024; Yang et al. 2024; Yim 2024; Zhang et al. 2024, 2023; Williams et al. 2023; Adams et al. 2023; Eguchi et al. 2021; Han et al. 2018; Kandlhofer et al. 2016; Lee et al. 2021
3	Assessment of student knowledge (n = 6)	Andries and Robertson 2023; Cai 2024; Chai et al. 2021; Dai et al. 2020; Mertala et al. 2022; Wu et al. 2024

Table 4 The GeST mapping of the scientific literature

Proposition	*Generic*	*Situated*	*Transformative*
AI literacy is …	n = 35	n = 1	n = 8
AI literacy is important because …	n = 22	n = 10	n = 11
AI literacy is taught and learned through …	n = 22	n = 10	n = 4
	n = 79	**n = 21**	**n = 23**

which introduce and evaluate different teaching activities in middle school classrooms (Eguchi 2023; Henry et al. 2021; Petridis et al. 2024; Williams et al. 2024; Yu and Huang 2024), in the evaluations of AI education curricula (Han et al. 2018; Höper and Schulte 2024; Moreno-León et al. 2024; Zhang et al. 2024), as well as in studies which assess what students know about AI (Cai 2024; Dai et al. 2020; Mertala et al. 2022; Wu et al. 2024). The teaching activities often follow a similar pattern of argument: (1) AI literacy is essential for students of all ages, (2) we developed activities and/or curricula to teach key AI concepts, (3) after evaluating the impact of our intervention, we found that students gained a better understanding of AI, and (4) our intervention enhances students' AI literacy. Rather than clarifying what AI literacy is, or perhaps because no consensus exists around what AI literacy should be, many of the studies focus on the effectiveness of their interventions, such as specially developed software for teaching about recommendation systems (Zhou et al. 2024a), robots designed to help understand AI concepts (Williams et al. 2024; Eguchi 2023), or using digital story-telling as a pedagogical approach (Ng et al. 2022b). In a similar vein, such patterns are also found in papers which develop and/or evaluate AI curricula for middle schools, as exemplified by Zhang et al. (2024):

> The results show that students in the experimental group developed a deeper understanding of AI concepts and more positive attitudes toward AI and its impact on future careers after the curriculum than those in the comparison group. This shows that the teacher-led classroom implementation successfully equipped students with a conceptual understanding of AI. Students achieved significant gains in recognising how AI is relevant to their lives and felt empowered to thrive in the age of AI (p. 23318).

What emerges here is not only that AI literacy consists of measurable skills, but also that it involves having 'a more positive attitude towards AI' and feeling 'empowered to thrive in the age of AI' (Appendix 1, Table 5).

A situated perspective (n = 1), which views AI literacy as an authentic and fundamentally collective activity emerging in specific contexts, is only reflected in the study by Williams et al. (2023). Their study, which also includes elements that can be attributed to a generic perspective, describes an AI curriculum that effectively engaged students in learning about and critiquing AI artifacts, based on three design principles: active learning, embedded ethics, and low barriers to access, aiming to meet the diverse needs of students from various backgrounds. Eight additional studies (n = 8) were seen as transformative

within this proposition as they describe AI literacy as practices that aim at transforming both individuals and society, offering teaching activities and principles designed to empower individuals to critically engage with AI and challenge the status quo. For example, Schaper and colleagues (2020) conducted a study with primary school students from diverse backgrounds, aiming to explore educational strategies for discussing the intersection of cultural diversity and AI risks. They used food practices as a conversational tool to engage children in discussions on ethical problems about AI, gender inequality, and how traditions shape norms and rules in our social life. Another study (Bilstrup et al. 2022) explores different design-oriented concepts for understanding machine learning in a critical way. Their suggested principles favoured reflexive engagement with the design of machine learning models over merely demonstrating their functionality. This meant that students were encouraged to engage not just with the technical components of machine learning models, but also with their design, the rationales behind their construction, and the context of their use and deployment. A final example is the analysis of AI ethics guidelines in policy reports by Adams and colleagues (2023) in which AI literacy emerges as an ethical principle, aligning it with a more transformative perspective.

AI literacy is important because …

Although less visible in the literature, the second proposition – why AI literacy is important – is also predominantly shaped by a generic perspective (n = 22). Studies which fall into this perspective tend to motivate the introduction of AI literacy in middle school education, when students start to think about their future careers (Yang et al. 2024) both to prepare a flexible and competitive workforce and introduce more AI technologies in middle school classrooms. For instance, Kandlhofer et al. (2016: 1) state that 'sound knowledge about AI and the principles of computer science will be of vast importance for future careers in science and engineering' and that 'Looking towards the near future, jobs will largely be related to AI'. Similarly, Lee and colleagues (2021: 191) argue that the lack of awareness of AI in everyday life may 'limit [youth's] interest in pursuing learning trajectories that lead toward careers in these fields'. Statements like these not only assume that AI is something fundamentally neutral, objective, and apolitical, but also that knowledge about AI leads to a globally flexible workforce, making a nation more competitive, as expressed by Zhang et al. (2024:23, 318): 'In the past decade, artificial intelligence (AI) is reshaping the industrial landscape and bringing tremendous economic benefits: 50% of companies that embrace AI over the next five to seven years have the potential to double their cash flow.' Although often discussed as something that comes with both opportunities and risks, a generic perspective generally promotes AI as something that can inherently improve education, as expressed by, for example, Eguchi (2021: 76): 'AI has the potential to improve the educational system by using data to promote equity and high-quality education in developing countries'.

About 20 papers (n = 21) are evenly distributed between the situated and the transformative perspective, and argue for AI literacy as a way to equip middle school students with context specific AI competencies to use AI, and to empower them in ways that ultimately lead to emancipatory outcomes. To exemplify, Ali et al. (2023: 15, 982) introduced a new game-based learning approach to instruct children about the societal

impact of AI systems using a card game. The game was designed to 'spark reflective conversation on everyday AI systems and raise awareness about their societal implications'. The transformative perspective extended not only to the empowerment of students but also to that of teachers. Another study by Ali et al. (2021) introduced middle school students to generative AI techniques, using the technology both as a creative tool and a means to foster critical discussions about its societal and ethical implications. In this context, empowering students involved helping them recognise that, as current users and future developers or policymakers, they could influence how deep fakes are implemented. As part of developing a data awareness framework, Höper and Schulte's study (2024: 492) aimed to 'empower students to navigate in a data-driven world, make informed and reflected decisions and actively manage these data flows'. The study suggests that a higher level of data awareness can help students reflect on their role in interactions with data-driven digital artifacts and, consequently, adopt a more critical stance toward these in their daily lives.

AI literacy is taught and learnt through …

The third proposition, how AI literacy is taught and learned, is the least reflected on and again dominated by the generic (n = 22) perspective. This is reflected in the way the teaching material focused on understanding specific technological aspects of AI and where it is applied (e.g., Drivas and Doukakis 2022; Henry et al. 2021; Pearce et al. 2023) but is also evident in papers which focus on curricula development (Han et al. 2018; Kandlhofer et al. 2016; Zhang et al. 2023). For example, in 2016, Kandlhofer and colleagues developed an AI literacy curriculum consisting of modules tailored for different age groups and educational levels, ranging from kindergarten through to university. Each module then addressed fundamental AI concepts such as graphs and data structures, sorting, and problem-solving through search. The modules employed various learning methods and techniques, including discovery learning, inquiry-based learning, collaborative learning, problem-based learning, project-based learning, story-telling, and peer teaching, or combinations of these methods.

The situated perspective is reflected in papers (n = 10) where the curricula and teaching materials were developed based on contextual preconditions (e.g. Breideband et al. 2023; Kahila et al. 2024; Ng et al. 2022b; Schaper et al. 2020). The study by Kahila et al. (2024) presented a gamified classroom activity designed to teach young learners fundamental concepts and principles of social media. Developed collaboratively by researchers, software developers, and schoolteachers, the activity engaged students in analysing data traces to construct and iteratively refine detailed profiles of a mystery online person. On evaluation, the game was seen to have developed 'an understanding of the digital footprints typical of online interactions'. The study by Ali et al. (2021) on teaching middle school students about generative AI provides an example of how AI literacy education can encompass generic elements that lead to measurable insights in individuals while adhering to situated content with the goal of empowering students to engage in AI practices. The teaching material consisted of three parts. In the first part, students were expected to gain an understanding of the technology. In the second part, they expanded on this knowledge to become aware of potentially harmful applications, using examples of disinformation relevant in a school context. In the third part, students applied their

knowledge by taking on the role of policymakers. The transformative perspective was seen in the studies of e.g. Bilstrup et al. (2022), who recommended that teachers create scenarios to reveal the complexity of the machine learning process and explore its limitations without overwhelming students with technical details. As a final example, the work of Di Paola and colleagues (2024: 2) presented the 'AI and Human Rights curriculum' that consists of seven lessons covering human rights, the impact of technology on our rights, and the Blueprint for an AI Bill of Rights. Situated in the US context, the study is clearly steered towards an emancipatory vision of AI literacy:

> Considering the influence of AI on our rights, we must prioritise AI literacy as an essential aspect of responsible citizenship. Not only are there parallels between the Bill of Rights and the AI Bill of Rights from an individual rights perspective, but AI also has the potential to be a powerful tool for or against shared civic goals, including safety and equity for all. In civics education, we hope AI can be another entry point for students to understand civic rights and the U.S. legislative process. (DiPaola et al. 2024: 2)

This suggests not only that the way AI should be taught and learned is context-dependent, but also that what constitutes responsible AI use, in the context of future citizenship, remains a widely open question.

Concluding Discussion

This integrated review has showed that AI and AI literacy are increasingly influencing the conversation about the future(s) of postdigital education (Buch et al. 2024), and that postdigital classrooms are sites where this future comes into being. We aimed to critically explore the current state and perspectives of AI literacy in the scientific literature focusing on middle school students, a group already exposed to AI in their daily lives and identified as particularly vulnerable and in need of proper AI guidance (UNICEF 2021). The rationale behind this focus was to understand not only what this specific age group should learn about AI but also how, and the potential outcomes they could achieve with such knowledge. The review of 44 papers published up to and including 2024 focusing on AI literacy related to middle school students demonstrates how little in-depth discussion of AI literacy exists within the scientific literature, despite a surge of papers in recent years. Although referencing to the term AI literacy, the papers rarely discuss the term itself, turning AI literacy to what Suchman (2023: 3) describes as a 'floating signifier' – a term intentionally kept vague to preserve its influence. By methodically analysing the literature through a holistic literacy framework, this study contributes to unpacking some of the vagueness and hidden assumptions of the term.

A Literacy in the Making

In response to our first descriptive research question on the current state of AI literacy relevant for students 9 to 13-years-old, we found that there is a growing interest in AI literacy that spans many parts of the world. Given the exponential growth of papers in recent years,

we expect this interest to grow. Currently, the dominance of contributions from the US, particularly from researchers at MIT, suggests that the field is still emerging and primarily led by individuals within geographically specific research groups. These researchers often work within specific research traditions, in this specific case with hands-on and exploratory approaches to learning (cf. Papert 1980). The studies tend to be solution-oriented, focusing on developing and evaluating analogue and digital teaching material for use in middle school classrooms. The focus on hands-on activities and project-based learning as a prominent feature in AI education has also been noted elsewhere (Lee and Kwon 2024; Sperling et al. 2024), and reflects the materiality inherent of emergent literacies now being introduced in postdigital classrooms. This technically oriented approach on AI literacy also mirrors an *engineering approach* to societal and educational challenges (cf. Macgilchrist et al. 2023) in which AI literacy is proposed as a technical solution to a different set of problems, such as getting a more skilful and competitive workforce and citizens who can adopt new technologies. As there is still limited understanding of, and no consensus on, what AI literacy *is* and why it should be integrated in compulsory schooling, this is an important albeit concerning finding. Thus, digital artefacts function both as an 'infrastructural backbone' (Decuypere et al. 2025) of AI literacy and, as a means, the literacy introduces AI technologies into postdigital classrooms. These technologies, and how they are designed, reflect global and national politics as well as commercial interests (Forsler et al. 2024) and shape what AI is and how AI literacy can be taught (Druga et al. 2022).

A Narrow Conceptualisation

The ways AI literacy is conceptualised has notable implications for how middle school students themselves understand AI, and how education about and with AI will prepare these students for a society where AI is ubiquitous. This motivated our second research question: What perspectives of AI literacy emerge in the literature? By adapting Lupton and Bruce's (2010) GeST framework, we were able to identify different AI literacy perspectives (generic, situated, and transformative) in relation to what AI literacy is, why it is important, and how it is taught and learnt. In line with Velander et al. (2024), our findings suggest that AI literacy is mainly concerned with a technological understanding and individual use that can be assessed, thereby reflecting a generic perspective. One explanation could be that many of the initiatives related to AI education and AI literacy thus far stem from the field of computer science education, which traditionally and naturally focuses on explaining, for example through the development of teaching materials, how AI technology works. Yet another explanation, is the political focus of preparing a flexible and competitive workforce in order to serve imagined economic and political outcomes. Here, the arguments that all levels of schooling need to teach AI literacy tend to remind us of a 'space race' mentality (cf. Mahanta 2025) emphasizing the urgency of producing the most competent younger generation. In other words, AI literacy that is underpinned by a generic perspective alone risk to present AI as something that emerges uniformly in different contexts. It also risks to overlook the fact that AI is a socio-technical phenomenon shaped by the convergence of multiple complex factors (Bearman et al. 2023). These include, among others, the advancement of multifaceted AI technologies in controlled research environments (Holmes and Tuomi 2022), the growth of a multi-million-dollar EdTech industry (Gulson and

Witzenberger 2022; Williamson 2020, 2024), and the increasing integration of AI within educational policy and governance (Holmes et al. 2022; Linderoth et al. 2024). Moreover, the tendency to simplify AI often obscures significant ethical concerns and issues, such as the exploitation of underpaid labour in the tech industry and the precarious working conditions faced by many workers engaged in repetitive tasks (Casilli et al. 2024; Tan and Cabato 2023). We argue, therefore, that for middle school students to interact with AI responsibly and productively, a certain level of theoretical knowledge and skills related to AI is essential. However, this alone is insufficient for an understanding of ethical, political, and economic dimensions that underpin these technologies, as well as for critically using AI in ways that are democratic and fair (Mannila et al. 2025).

Towards a Holistic AI Literacy in Postdigital Classrooms

Similarly to how the critical literacy movement has criticised notions of literacy as emphasising the individual in favour of the social (e.g. Janks 2013), the results from this study highlight the need to explore and strengthen the social practices and emancipatory dimensions of AI literacy as it emerges in postdigital classrooms (cf. Pangrazio and Sefton-Green 2020). The situated aspects are important to create personal relevance for students, and to recognise that knowledge is more than subjective, but also collective. The transformative perspective again helps students to have a critical mindset and asks questions related to, for example, who generated the data and developed the specific AI technology and for what purposes. Although we found studies that reflect both situated and transformative perspectives (e.g. Burris et al. 2024; Williams et al. 2023), these perspectives are less common for AI literacy to truly support middle school students' agency and imagining of alternative futures and narratives. At the same time, our review also demonstrates that it is possible even for young learners in classroom settings to achieve both a situated and a transformative perspective using different kinds of strategies and approaches. Such strategies include e.g. teaching activities that require students to ask questions about power, bias, and purpose in AI systems, using real world scenarios like facial recognition or deep fake to understand the societal impact of AI, and to engage students in critical or speculative design activities that offer imagine alternative futures with AI (Bilstrup et al. 2022; DiPaola et al. 2024; Schaper et al. 2020). Moving towards a holistic AI literacy also implies supporting middle school teachers in understanding AI's societal dimensions (Lee et al. 2022). From this outset, a holistic approach encompassing generic, situated, and transformative perspectives on AI could empower students of all ages to explore their own identities, critically engage with the world around them, and enter into meaningful dialogue with it (cf. Biesta 2021).

From a methodological perspective, the adapted GeST framework has provided the review with analytical rigor, showing potential to capturing and analysing AI literacy. Future studies can benefit from using the adapted framework to further unpack AI literacy as a dynamic and evolving construct that manifests differently across various parts of the world. Furthermore, the framework can serve as a useful tool for designing AI curricula that foster critical thinking and active citizenship among middle school students, shaping not only their decisions, worldviews, and agency, but also enriching their overall educational experiences and proposing alternative future visions for postdigital classrooms.

Appendix 1

Table 5 Mapping of included papers

Paper	Theme	G	S	T
Zhou, X., Y. Zhou, Y. Gong, Z. Cai, A. Qiu, Q. Xiao, A.N. Antle, och Z. Bai. (2024a). 'Bee and I need diversity!' Break Filter Bubbles in Recommendation Systems through Embodied AI Learning	Teaching materials/activities	1		
				1
			1	
Zhou, X., Xiong, P., Xiao, Q., & Bai, Z. (2024b). OptiDot: An Optical Interface for Children to Explore Dot Product and AI Recommendation	Teaching materials/activities	1		
		1		1
			1	
Zhang, H., Lee, I., & Moore, K. (2024). An Effectiveness Study of Teacher-Led AI Literacy Curriculum in K-12 Classrooms	Curriculum development	1		
		1		
		1		
Yu, W., & Huang, W. (2024). Design and Development of an AI Education Program for Young Learners: A Case Study	Teaching materials/activities	1		
		1		
		1		
Yim, I. H. Y. (2024). Artificial intelligence literacy in primary education: An arts-based approach to overcoming age and gender barriers	Curriculum development	1		
		1		
		1		
Yang, H., Zhang, D., Guo, W., & He, Y. (2024). Constructing a STEM Projects Design Framework for Middle School AI Courses: Developing Students' AI Literacy as Project Goals	Curriculum development	1		
		1		
		1		
Wu, D., Chen, M., Chen, X., & Liu, X. (2024). Analyzing K-12 AI education: A large language model study of classroom instruction on learning theories, pedagogy, tools, and AI literacy	Assessment of student knowledge	1		
		1		
		1		
Williams, R., Ali, S., Alcantara, R., Burghleh, T., Alghowinem, S., & Breazeal, C. (2024). Doodlebot: An Educational Robot for Creativity and AI Literacy	Teaching materials/activities	1		
		1		
		1		

Table 5 (continued)

Paper	Theme	G	S	T
Tseng, Y.-J., Yadav, G., Hou, X., Wu, M., Chou, Y.-S., Chen, C. C., Wu, C.-C., Chen, S.-G., Lin, Y.-J., Liao, G., & Koedinger, K. R. (2024). ActiveAI: The Effectiveness of an Interactive Tutoring System in Developing K-12 AI Literacy	Teaching materials/activities	1		
		1		
		1		
Simbeck, K., & Kalff, Y. (2024). Understanding how Computers Learn: AI Literacy for Elementary School Learners	Curriculum development	1		1
			1	
		1		
Petridis, P., Benson, V., Garibyan, M., Meireles, G., Carpov, A., Dimitrakopoulou, D., & Teles, M. (2024). Adventure in AI Project (2 AI): Promoting AI Knowledge for Kids Aged 7–12 Using Gaming	Teaching materials/activities	1		
		1		
		1		
Moreno-León, J., Vasco-González, M., Román-González, M., & Robles, G. (2024). Investigating the Impact of Programming Activities on Computational Thinking and AI Literacy in Spanish Schools	Curriculum development	1		
		1		
		1		
Kim, W. J., & An, S. (2024). Shifting perceptions of AI: An integrated learning program in the middle school setting	Curriculum development	1		
			1	
			1	
Kahila, J., Vartiainen, H., Arkko, E., Lin, A., Pope, N., & Tedre, M. (2024). Enhancing Understanding of Data Traces and Profiling Among K-9 Students Through an Interactive Classroom Game	Teaching materials/activities			1
			1	
			1	
Höper, L., & Schulte, C. (2024). The data awareness framework as part of data literacies in K-12 education	Curriculum development	1		
				1
		1		
Hollands, F. M., Dipaola, D., Breazeal, C., & Ali, S. (2024). AI Mastery May Not Be For Everyone, But AI Literacy Should Be	Curriculum development	1		
		n/a	n/a	n/a
		1	1	

Table 5 (continued)

Paper	Theme	G	S	T
DiPaola, D., Aidinoff, M., Marzilli, A., Berman, E., Champigny, M., Gundy, N., Yezzi, S., & Breazeal, C. (2024). AI literacy as civic literacy: A case study of the "AI and human rights" curriculum for middle school students	Curriculum development			1
			1	
				1
Choi, J.-I., Yang, E., & Goo, E.-H. (2024). The Effects of an Ethics Education Program on Artificial Intelligence among Middle School Students: Analysis of Perception and Attitude Changes	Curriculum development	1		
		1	1	
		1		
Chiu, T. K. F., Ahmad, Z., Ismailov, M., & Sanusi, I. T. (2024). What are artificial intelligence literacy and competency? A comprehensive framework to support them	Curriculum development	1		
		1		
		n/a	n/a	n/a
Cai, L. (2024). Realities vs expectations: Children's perception and imagination of AI	Assessment of student knowledge	1		
		1		
		n/a	n/a	n/a
Burriss, S. K., Hutchins, N., Conley, Z., Deweese, M. M., Doe, Y. J., Eeds, A., Villanueva, A., Ziegler, H., & Oliver, K. (2024). Redesigning an AI bill of rights with/for young people: Principles for exploring AI ethics with middle and high school students	Curriculum development			1
			1	
				1
Zhang, H., Lee, I., Ali, S., DiPaola, D., Cheng, Y., & Breazeal, C. (2023). Integrating Ethics and Career Futures with Technical Learning to Promote AI Literacy for Middle School Students: An Exploratory Study	Curriculum development	1		
		1		1
		1		
Williams, R., Ali, S., Devasia, N., DiPaola, D., Hong, J., Kaputsos, S. P., Jordan, B., & Breazeal, C. (2023). AI +Ethics Curricula for Middle School Youth: Lessons Learned from Three Project-Based Curricula	Curriculum development	1	1	
		1	1	
Pearce, K., Alghowinem, S., & Breazeal, C. (2023). Build-a-Bot: Teaching Conversational AI Using a Transformer-Based Intent Recognition and Question Answering Architecture	Teaching materials/activities	1		
			-	
		1		

Table 5 (continued)

Paper	Theme	G	S	T
Kong, S.-C., & Yang, Y. (2023). Designing and Evaluating an Attention-Engagement-Error-Reflection (AEER) Approach to Enhance Primary School Students Artificial Intelligence Literacy and Learning-to-Learn Skills: A Pilot Study	Teaching materials/activities	1		1
			1	
		1		
Breideband, T., Bush, J., Chandler, C., Chang, M., Dickler, R., Foltz, P., Ganesh, A., Lieber, R., Penuel, W. R., Reitman, J. G., Weatherley, J., & D'Mello, S. (2023). The Community Builder (CoBi): Helping Students to Develop Better Small Group Collaborative Learning Skills	Teaching materials/activities		n/a	
			n/a	
			1	
Eguchi, A. (2023). Machine Learning Application with AI-Robotics Tool—What We Learned from Pilot Studies	Teaching materials/activities	1		
				1
			1	
Andries, V., & Robertson, J. (2023). Alexa doesn't have that many feelings: Children's understanding of AI through interactions with smart speakers in their homes	Assessment of student knowledge			1
		1		
		n/a	n/a	n/a
Ali, S., Kumar, V., & Breazeal, C. (2023). AI Audit: A Card Game to Reflect on Everyday AI Systems	Teaching materials/activities			1
				1
			1	
Adams, C., Pente, P., Lemermeyer, G., & Rockwell, G. (2023). Ethical principles for artificial intelligence in K-12 education	Curriculum development			1
				1
		n/a	n/a	n/a
Mertala, P., Fagerlund, J., & Calderon, O. (2022). Finnish 5th and 6th grade students' pre-instructional conceptions of artificial intelligence (AI) and their implications for AI literacy education	Assessment of student knowledge	1		
			1	
		n/a	n/a	n/a
Drivas, T., & Doukakis, S. (2022*).* Introducing the Fundamentals of Artificial Intelligence to K-12 Classrooms According to Educational Neuroscience Principles	Teaching activities/materials	1		
		1		
		1		

Table 5 (continued)

Paper	Theme	G	S	T
Ng, D.T.K**,** Luo, W., Chan, H. M. Y., & Chu, S. K. W. (2022). Using digital story writing as a pedagogy to develop AI literacy among primary students	Teaching activities/materials	1		
		1		
			1	
Bilstrup, K.-E. K., Kaspersen, M. Hø., Assent, I., Enni, S., & Petersen, M. G. (2022). From Demo to Design in Teaching Machine Learning	Teaching materials/activities	1		
		1		
				1
Voulgari, I., Zammit, M., Stouraitis, E., Liapis, A., & Yannakakis, G. (2021). Learn to Machine Learn: Designing a Game Based Approach for Teaching Machine Learning to Primary and Secondary Education Students	Teaching activities/materials	1		
			1	
Henry, J., Hernalesteen, A., & Collard, A.-S. (2021). Teaching Artificial Intelligence to K-12 Through a Role-Playing Game Questioning the Intelligence Concept	Teaching activities/materials	1		
				1
		1		
Eguchi, A., Okada, H., & Muto, Y. (2021). Contextualizing AI Education for K-12 Students to Enhance Their Learning of AI Literacy Through Culturally Responsive Approaches. *KI—Kunstliche Intelligenz, 35*(2), 153–161	Curriculum development	1		
			1	
			1	
Lee, I., Ali, S., Zhang, H., Dipaola, D., & Breazeal, C. (2021). Developing Middle School Students' AI Literacy	Curriculum development	1		
		1	1	
			1	
Chai, C. S., Lin, P.-Y., Jong, M. S.-Y., Dai, Y., Chiu, T. K. F., & Qin, J. (2021). Perceptions of and Behavioral Intentions towards Learning Artificial Intelligence in Primary School Students	Assessment of student knowledge	1		
		1		
		1		
Ali, S., DiPaola, D., Lee, I., Sindato, V., Kim, G., Blumofe, R., & Breazeal, C. (2021). Children as creators, thinkers and citizens in an AI-driven future	Teaching activities/materials			1
				1
		1	1	

Table 5 (continued)

Paper	Theme	G	S	T
Schaper, M.-M., Malinverni, L., & Valero, C. (2020). Robot Presidents: Who should rule the world?: Teaching Critical Thinking in AI through Reflections upon Food Traditions	Teaching activities/materials			1
				1
			1	
Dai, Y., Chai, C.-S., Lin, P.-Y., Jong, M. S.-Y., Guo, Y., & Qin, J. (2020). Promoting students' well-being by developing their readiness for the artificial intelligence age	Assessment of student knowledge	1		
		1		1
		n/a	n/a	n/a
Han, X., Hu, F., Xiong, G., Liu, X., Gong, X., Niu, X., Shi, W., & Wang, X. (2018). Design of AI +Curriculum for Primary and Secondary Schools in Qingdao	Curriculum development	1		
		1		
		1		
Kandlhofer, M., Steinbauer, G., Hirschmugl-Gaisch, S., & Huber, P. (2016). Artificial intelligence and computer science in education: From Kindergarten to university	Curriculum development	1		
		1		
		1		

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Data Availability Research data can be shared upon request.

Declarations

Conflicts of interest All authors declare that they have no conflicts of interest.

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